

Summary

AI Robotics Engineer at AMTRAN TECHNOLOGY CO., LTD. with an M.S. in Electrical Engineering. Build robot control and learning systems across ROS 2, TM5S, NVIDIA Isaac Sim / Isaac Lab, MoveIt 2, cuMotion, and Isaac GROOT, integrating perception, data workflows, policy inference, and real-robot validation.

Experience

- AMTRAN TECHNOLOGY CO., LTD. | AI Robotics Engineer

May 2025 - Present | New Taipei, Taiwan

 - Built ROS 2 control and simulation environments connecting TM5S, NVIDIA Isaac Sim / Isaac Lab, MoveIt 2, and cuMotion for motion planning, collision handling, pick-and-place, and cable tasks.
 - Delivered Isaac GROOT N1.7 data conversion, model training, policy inference, and TM5S real-robot testing, integrating camera data, robot state / action, policy services, and robot commands.
 - Integrated RealSense, YOLO OBB, AprilTag, and OpenCV for 3D pose estimation, hand-eye calibration, TF conversion, and visual servoing in physical robot workflows.
 - Built Quest 2 teleoperation and imitation-learning recording workflows; automated task execution and data collection reached 3-4x manual operation in the same time window.
 - Independently delivered robotics software from requirements and architecture through Docker deployment, testing, on-site integration, and maintenance; cable-insertion success improved from 70% to 90% in a 10-trial test.
- ISCOM | Software Engineer Intern

Feb 2022 - Jun 2022 | Kaohsiung, Taiwan

 - Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Robotics & AI	ROS 2, TM5S, NVIDIA Isaac Sim / Lab, MoveIt 2, cuMotion, Isaac GROOT N1.7, VLA / Policy Inference, Imitation Learning
Perception & Data	OpenCV, YOLO OBB, AprilTag, RealSense, Hand-Eye Calibration, Visual Servoing, HDF5, Robot State / Action
Software	Python, C++, C# / .NET, PyQt, FastAPI, React / TypeScript, Docker, Linux, PostgreSQL / SQLite, Git

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
 - B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
 - Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place
- Portfolio: daniel-locy.github.io